

Authentication Guide

Project **multi-tenant-ecommerce**: NestJS + Prisma backend, **platform-admin** frontend on React + Redux

A step-by-step walkthrough: how authentication works in the project, what is implemented where on the backend and frontend, which concepts to know, what to remember and what to watch out for.

Repository	github.com/ShipaM/multi-tenant-ecommerce
Code state	commit c1d2eb9 from 2026-10-02 (default branch)
Covered	backend/src/auth , backend/src/users , backend/src/common , backend/prisma , platform-admin/src (store, lib, api, routes, pages)
Audience	Team members who work on or integrate with authentication

REMEMBER How to read this guide

Parts 1-4 give the overview and glossary. Parts 5-6 cover the backend and frontend layer by layer. Part 7 gives step-by-step scenarios (login, 2FA, refresh, logout, password change and reset). Parts 8-9 cover strengths and known gaps. Part 10 is a checklist and review questions.

Boxes: **REMEMBER** = key point, **WARNING** = pitfall, **IDEA** = improvement. File paths are given from the repository root.

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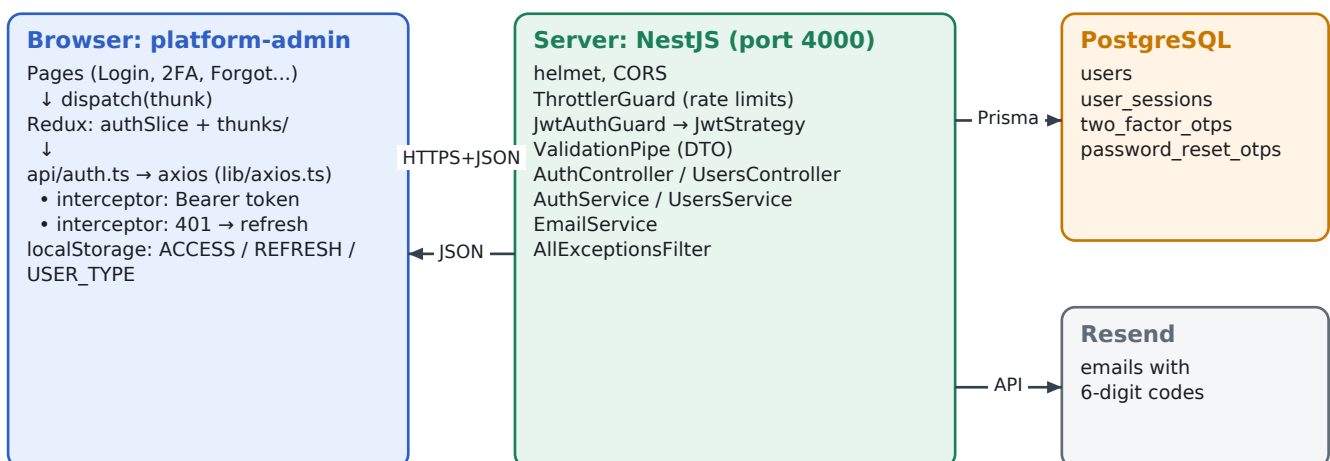
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1. The Big Picture

1.1. What the project consists of

Part	Technologies	Role in authentication
backend/ (port 4000)	NestJS 12 (CommonJS), Prisma 7 + PostgreSQL, passport-jwt, @nestjs/jwt, bcryptjs, @nestjs/throttler, helmet, class-validator, Resend (email), ua-parser-js, vitest	Verifies the password, issues and verifies tokens, stores sessions and one-time codes, sends emails with codes
platform-admin/ (port 3002)	React 19, Vite, react-router, Redux Toolkit, axios, shadcn/ui, sonner, moment, tests: vitest + Testing Library	Login forms, token storage, automatic token refresh, page protection, the "My account" page
PostgreSQL	tables <code>users</code> , <code>user_sessions</code> , <code>two_factor_otps</code> , <code>password_reset_otps</code>	The single source of truth about users and sessions
Resend	external email service	Delivers 6-digit codes (2FA and password reset)

1.2. Overall diagram



1.3. What authentication does in the project

- **Login by email + password** (`POST /auth/login`) issuing a token pair and creating a session.
- **Two-factor authentication (2FA)** with a code from an email: both at login and as a setting that can be enabled/disabled.
- **Access + refresh tokens** with automatic renewal on the frontend and refresh token **rotation** (reuse of an old refresh token revokes the session).
- **Server-side sessions** (`user_sessions` table): device list, logout of a specific device, logout of all other devices.
- **Logout** (`POST /auth/logout`) with immediate token invalidation.
- **Password change** (current password known) and **password recovery** (forgotten) via a code from an email; the reset token is single-use and a reset closes all sessions.
- **Profile** (`PUT /users/me`): name, email, phone, photo.
- **Brute-force protection**: request rate limits, a limit on code entry attempts, identical responses for non-existent emails.

WARNING About "multi-tenant": a clarification

The project name says multi-tenant, but the **authentication code has no concept of a "tenant" (store/organization) yet**: the DB schema has no tenants table and tokens carry no `tenantId`. There is only the `userId` field (`CUSTOMER`, `SELLER`, `PLATFORM_ADMIN`, `DELIVERY_AGENT`).

The `platform-admin` frontend is the platform owner's console (text on the login screen: "Run every store on your marketplace from one place"). Data isolation between tenants is the next stage; Part 9 notes what needs to be added for it.

2. Key Concepts

This is a glossary to come back to. The "Where in the project" column shows where each concept is used.

Basics

Concept	What it is	Where in the project
Authentication (authN)	Determine who is making the request: check the email and password, the code from the email.	<code>AuthService.login</code> , <code>JwtStrategy</code>
Authorization (authZ)	Decide what this user is allowed to do : roles, permissions, access to tenant data.	Not implemented in the project yet (see Part 9)
Session	A record saying "this user is logged in on this device". Makes it possible to revoke a login.	<code>user_sessions</code> table
Stateless / stateful	Stateless: the server stores nothing and trusts the token signature. Stateful: the server stores state (a session) and checks against it.	Hybrid: JWT + a session check in the DB on every request

Tokens

Concept	What it is	Where in the project
JWT	A string <code>header.payload.signature</code> . The payload is plain base64url (anyone can read it), the signature prevents forging it.	<code>JwtService.signAsync</code> / <code>verifyAsync</code>
Access token	Short-lived (15 min) token for API requests in the <code>Authorization: Bearer ...</code> header.	<code>signTokenPair</code> , <code>JwtStrategy</code>
Refresh token	Long-lived (30 days) token, used only to obtain a new token pair.	<code>AuthService.refresh</code> , <code>lib/axios.ts</code>
Refresh rotation	Each renewal issues a new refresh token and the old one stops working.	<code>refresh()</code> → <code>userSession.update</code>
Refresh token reuse detection	If an already replaced (old) refresh token is presented, the session is revoked: a stolen old token kills the session. Side effect: two tabs refreshing with the same old token may log the user out.	<code>AuthService.refresh</code> (<code>revokedAt</code>)
Single-use reset token	The reset token carries <code>passwordVersion</code> (the password change time); after the first reset the version changes, and reusing the token gives "Session is Expired".	<code>resetPassword</code> , <code>resetForgottenPassword</code>
Bearer	Token transport scheme: <code>Authorization: Bearer <token></code> .	interceptor in <code>lib/axios.ts</code>
Signing secret	A string of at least 32 characters used to sign JWTs. Each token type has its own.	<code>JWT_*_SECRET</code> in <code>.env</code>

Passwords, codes and hashes

Concept	What it is	Where in the project
Hash	An irreversible "fingerprint" of data. The password is stored in the DB only as a hash.	<code>passwordHash</code>
bcrypt	A slow, salted password hash; the <code>cost=10</code> parameter sets how slow it is.	<code>bcrypt.hash(pw, 10)</code>
HMAC-SHA256	A hash with a secret key. It hashes the refresh token before it is written to the DB.	<code>hashRefreshToken</code>
OTP	One-Time Password: a single-use 6-digit code, valid for 5 minutes, allows 5 attempts.	<code>two_factor_otps</code> , <code>password_reset_otps</code>
2FA	Second factor: besides knowing the password, the user must prove access to the mailbox.	<code>twoFactorEnabled</code>
Timing attack	Guessing from response time. Defense: constant-time comparison.	<code>timingSafeEqual</code> , dummy hash
Email enumeration	Finding out whether an email is registered from differences in responses.	identical responses in <code>login</code> , <code>forgotPassword</code>

Backend (NestJS)

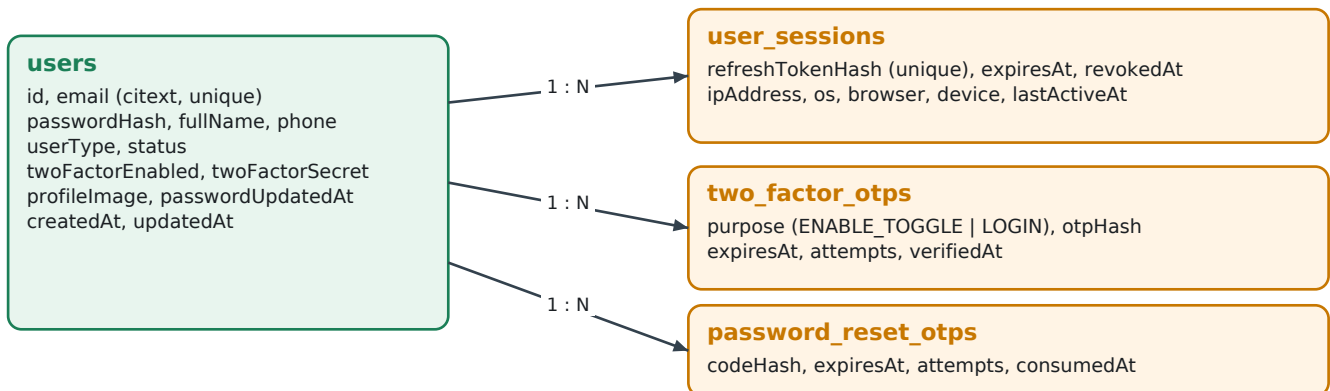
Concept	What it is	Where in the project
Module / Controller / Service	A module groups code; a controller receives the HTTP request; a service holds the logic.	<code>AuthModule</code> , <code>AuthController</code> , <code>AuthService</code>
DTO + ValidationPipe	A DTO is a class describing the request body; a pipe validates and cleans the data before the controller.	<code>LoginDto</code> , <code>dto/index.ts</code> (barrel), <code>main.ts</code>
Guard	Decides whether a request may reach the handler. Runs before pipes.	<code>JwtAuthGuard</code> , <code>ThrottlerGuard</code>
Passport Strategy	A rule for verifying the request credentials (here, the JWT from the header).	<code>JwtStrategy</code>
Decorator	<code>@Post</code> , <code>@UseGuards</code> , <code>@Throttle</code> , <code>@CurrentUser</code> : annotations/tools applied to methods and parameters.	<code>current-user.decorator.ts</code>
Exception Filter	A single place where errors are turned into HTTP responses.	<code>AllExceptionsFilter</code>
Prisma	ORM: the schema is described in <code>schema.prisma</code> , producing a typed client.	<code>PrismaService</code>

Frontend (React)

Concept	What it is	Where in the project
Redux Toolkit slice	State store + reducers for one area (auth); the thunks live separately.	<code>store/auth/authSlice.ts</code>
<code>createAsyncThunk</code>	An async action with pending / fulfilled / rejected states.	<code>store/auth/thunks/</code> (<code>fetchLogin</code> , <code>fetchMe...</code>)
<code>unwrap()</code>	Turns the thunk result into a plain promise: success → value, error → <code>throw</code> .	<code>dispatch(...).unwrap()</code> in pages
Interceptor (axios)	Intercepts every request/response: adds the token, catches 401.	<code>lib/axios.ts</code>
Protected route	A wrapper route that blocks access without a token.	<code>ProtectedRoute.tsx</code>
<code>localStorage</code>	Browser storage; survives reloads and is readable by any JS on the page.	<code>lib/storage.ts</code>
Open redirect	An attack via a parameter like <code>?redirect_uri=https://evil.com</code> .	<code>isSafeRedirectPath</code>

3. Database (Prisma)

File: `backend/prisma/schema.prisma`. All related tables reference `users` and have `onDelete: Cascade`: when a user is deleted, their sessions and codes are deleted automatically.



3.1. Tables and fields worth understanding

Table.field	Meaning
<code>users.email</code>	Type <code>citext</code> (PostgreSQL extension): case-insensitive comparison, plus <code>@unique</code> . <code>Admin@x.com</code> and <code>admin@x.com</code> are the same user.
<code>users.passwordHash</code>	bcrypt hash of the password. Never returned to the client (<code>omit</code> is used in responses).
<code>users.userType</code>	User type (enum). Currently only stored and put into the token ; access is not checked against it.
<code>users.status</code>	<code>ACTIVE</code> / <code>SUSPENDED</code> / <code>INACTIVE</code> . Only <code>ACTIVE</code> users can log in and use the API (checked at login, at refresh and on every request).
<code>users.twoFactorEnabled</code>	Whether 2FA is enabled. If yes, a code from an email is required after the password.
<code>users.twoFactorSecret</code>	Reserved for TOTP apps (Google Authenticator). Currently unused : codes are sent by email.
<code>users.passwordUpdatedAt</code>	When the password was last changed (shown on the account page: "changed ... ago"). Also serves as the "password version" in the reset token, making it single-use.
<code>user_sessions</code> (row)	One login on one device. Its <code>id</code> goes into the access token as <code>sid</code> .
<code>user_sessions.refreshTokenHash</code>	HMAC hash of the current refresh token (the token itself is not stored).
<code>user_sessions.expiresAt</code>	Session lifetime = the refresh token <code>exp</code> . Updated on every refresh.
<code>user_sessions.revokedAt</code>	If not <code>null</code> , the session is revoked and its tokens stop working immediately.
<code>user_sessions.lastActiveAt</code>	Last activity: updated on refresh and on authenticated requests (at most once per 60 s). The device list shows only sessions that are neither revoked nor expired.
<code>user_sessions.os / browser / device / ipAddress</code>	Parsed <code>User-Agent</code> (ua-parser-js) and IP, used for the device list.
<code>two_factor_otps.purpose</code>	<code>LOGIN</code> (code for login) or <code>ENABLE_TOGGLE</code> (code for enabling/disabling 2FA). A code for one purpose does not work for the other.
<code>*_otps.otpHash / codeHash</code>	bcrypt hash of the 6-digit code: the code itself is not in the DB.
<code>*_otps.attempts</code>	Counter of wrong attempts. After 5: "Too many otp attempts".
<code>two_factor_otps.verifiedAt</code> , <code>password_reset_otps.consumedAt</code>	When the code was successfully used: makes the code single-use.

3.2. Migration history

Migration	What it added
20260907 <code>users_session_tables</code>	<code>citext</code> , enums, <code>users</code> , <code>user_sessions</code> , <code>password_reset_otps</code>
20260910 <code>profile_image</code>	<code>users.profile_image</code>
20260917 <code>two_factor_otp_and_expiry_indexes</code>	<code>two_factor_otps</code> , indexes on <code>expires_at</code>
20260918 <code>two_factor_otp_purpose</code>	<code>purpose</code> field (<code>LOGIN</code> / <code>ENABLE_TOGGLE</code>)
20260922 <code>user_agent</code>	<code>os</code> , <code>device</code> , <code>browser</code> in the session
20260924 <code>added_password_updated_at_field</code>	<code>users.password_updated_at</code>
20260929 <code>password_reset_otp_attempts</code>	<code>attempts</code> in <code>password_reset_otps</code>

REMEMBER The main idea of the schema

User $\neq$ session $\neq$ token. A token is a "pass", a session is an "entry in the pass log". The pass is valid only while the log entry is neither revoked nor expired, so logout and blocking take effect immediately, not after 15 minutes.

4. Four Kinds of JWT

The project signs **four different tokens**, each with its own secret and lifetime. Separate secrets ensure a token of one kind cannot be passed off as another.

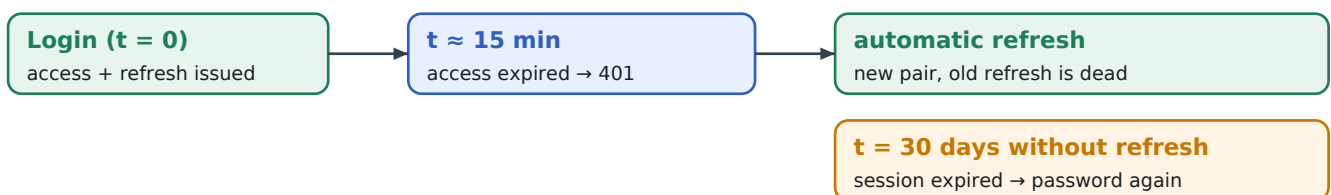
Token	Secret (.env)	TTL	Payload	Where stored	Purpose
Access	JWT_ACCESS_SECRET	15m	userId, email, userType, sid	Client: localStorage. Server: nowhere	Access to protected endpoints (Bearer)
Refresh	JWT_REFRESH_SECRET	30d	userId, sessionId	Client: localStorage. Server: only the HMAC hash in user_sessions	Obtain a new token pair
2FA	JWT_2FA_SECRET	15m	userId	Client: router location.state . Server: nowhere	twoFactorToken : proves "the password is already verified" while waiting for the emailed code
Reset	JWT_RESET_SECRET	5m	userId, passwordVersion	Client: location.state . Server: nowhere	resetToken : the right to set a new password after entering the code; single-use (passwordVersion = passwordUpdatedAt at issue time)

TTL is the default lifetime. Values can be changed in **.env** (format **15m**, **7d**, **3600s**, validated by a regular expression in **env.validation.ts**).

4.1. Why two tokens (access + refresh)

- The access token is sent with **every** request, so it is more likely to leak. That is why it lives 15 minutes: a stolen token quickly becomes useless.
- The refresh token is used **rarely** (about once per 15 minutes and only on **/auth/refresh**). It lives long so users do not have to enter their password every 15 minutes.
- On the server the refresh token is bound to a session: it can be revoked (**revokedAt**) and is replaced on every use; presenting an old refresh token again revokes the whole session.

4.2. Login timeline



REMEMBER A JWT is signed, not encrypted

Anyone can decode a token payload (it is just base64url). The signature protects against **forgery**, not against **reading**. So passwords and secrets must never go into a token. The access token here contains **userId**, **email**, **userType** and **sid**, which is fine.

Try it: open DevTools → Application → Local Storage, copy **ACCESS_TOKEN** and run in the console `JSON.parse(atob(token.split('.')[1].replace(/-/g, '+').replace(/_/g, '/')))` to see the payload and the **exp** field.

5. Backend by layers (NestJS)

5.1. The request path through the server

Order matters: if a request is rejected at an early step, it never reaches the later ones.

1. **helmet** (`main.ts`) — adds protective HTTP headers to every response.
2. **CORS** (`main.ts`) — allows requests only from origins listed in **CORS_ORIGINS** (for example `http://localhost:3002`). If the variable is not set, CORS is disabled.
3. **ThrottlerGuard** (global, `app.module.ts`) — request rate limit: 60 per minute by default; sensitive routes have their own, stricter limits (`@Throttle`).
4. **JwtAuthGuard** (only on protected routes via `@UseGuards`) — checks the token and the session.
5. **ValidationPipe** (global) — validates the request body against the DTO.
6. **Controller → Service → Prisma** — business logic and the database.
7. **AllExceptionsFilter** — turns any error into a single JSON shape `{ statusCode, message, error, path, timestamp }`; unhandled errors are returned as "Internal server error" (without a stack trace), while the stack is written to the log.

5.2. `main.ts`: global settings

```
app.use(helmet());
app.useGlobalPipes(new ValidationPipe({
  whitelist: true,           // strip fields that are not in the DTO
  forbidNonWhitelisted: true, // and if extra fields are present, return 400
  transform: true,           // turn JSON into an instance of the DTO class
  transformOptions: { enableImplicitConversion: true },
}));
app.useGlobalFilters(new AllExceptionsFilter());
if (corsOrigins) app.enableCors({ origin: [...], credentials: true });
void bootstrap(); // CommonJS: no top-level await, so the promise is started explicitly
```

REMEMBER whitelist + forbidNonWhitelisted = protection against mass assignment

In `UsersService.update`, Prisma receives `data: { ...dto }`. This is safe only because the pipe does not let extra fields into the DTO. Otherwise a client could send `{ "userType": "PLATFORM_ADMIN" }` and escalate its own privileges.

- **Module system: CommonJS.** `"type": "module"` was removed from `backend/package.json`, so the **nodenext** build now emits CJS: imports have no `.js` suffix (`./auth.service`, `./dto/index`), the Prisma generator has `moduleFormat = "cjs"`, and `main.ts` ends with `void bootstrap();` (no top-level await).
- Vitest configs are `vitest.config.mts` / `vitest.config.e2e.mts`; the seed imports `../src/generated/prisma/client`.

5.3. `app.module.ts`, configuration and `.env`

- `ConfigModule.forRoot({ isGlobal, cache, validate: validateEnv })` — environment variables are validated **at startup**: if a secret is missing or shorter than 32 characters, the application will not start (`env.validation.ts`).

- `ThrottlerModule.forRoot + APP_GUARD: ThrottlerGuard` — a global limit of 60 requests/min.
- `GlobalPassportModule` — registers Passport with `jwt` as the default strategy.
- Each token kind has its own secret. The reset token (`JWT_RESET_SECRET`) carries `{ userId, passwordVersion }`, where `passwordVersion` is `users.password_updated_at` in epoch ms at the time of issue — this is what makes it single-use (section 5.7).

Variable	Required?	Value
<code>DATABASE_URL</code>	yes	PostgreSQL connection string
<code>JWT_ACCESS_SECRET</code> , <code>JWT_REFRESH_SECRET</code> , <code>JWT_2FA_SECRET</code> , <code>JWT_RESET_SECRET</code>	yes	four different random strings, each at least 32 characters
<code>JWT_ACCESS_EXPIRES_IN</code> / <code>JWT_REFRESH_EXPIRES_IN</code> / <code>JWT_2FA_EXPIRES_IN</code> / <code>JWT_RESET_EXPIRES_IN</code>	no	default <code>15m</code> / <code>30d</code> / <code>15m</code> / <code>5m</code>
<code>RESEND_API_KEY</code> , <code>RESEND_FROM_EMAIL</code>	yes	API key and sender address for emails
<code>CORS_ORIGINS</code>	no	comma-separated list of origins; without it CORS is disabled
<code>PORT</code> , <code>NODE_ENV</code>	no	default <code>4000</code> and <code>development</code>

5.4. Authentication endpoints

Method and path	Protection	Limit	Body	What it does
POST /auth/login	public	10	email, password	Checks the password. Returns tokens or a 2FA requirement
POST /auth/refresh	public	20	refreshToken	Rotation: a new token pair; a stale (already rotated) token revokes the session
POST /auth/logout	Bearer	60	—	Revokes the current session
GET /auth/me	Bearer	60	—	Profile without passwordHash and twoFactorSecret
POST /auth/2fa-generate-otp	Bearer	5	—	Sends a code for enabling/disabling 2FA
POST /auth/2fa-verify-enable	Bearer	5	otp	Verifies the code and toggles twoFactorEnabled
POST /auth/2fa-verify-login-otp	public	5	otp, twoFactorToken	Completes a 2FA login (identity is proven by twoFactorToken), issues tokens
GET /auth/sessions	Bearer	60	—	Sessions that are not revoked and not expired, newest activity first
POST /auth/sessions/:id/revoke	Bearer	60	—	Revoke a single session
POST /auth/sessions/revoke-others	Bearer	60	—	Revoke all except the current one
POST /auth/forgot-password	public	5	email	Sends a reset code by email
POST /auth/forgot-password/verify-otp	public	5	email, otp	Verifies the code (VerifyForgotPassword0tpDto) → resetToken
POST /auth/reset-password	public	5	resetToken, password	Sets a new password (the right comes from resetToken, which works only once); revokes all sessions
PUT /users/me	Bearer	60	fullName, email, phone, profileImage	Update the profile
PUT /users/change-password	Bearer	60	currentPassword, password	Change the password

"Limit" is requests per minute; 60 is the global default. By default Throttler counts requests per IP.

5.5. AuthService: what each method does

Method	Essence
<code>login(email, password, ctx)</code>	Looks up the user; always runs <code>bcrypt.compare</code> (for a non-existent user, against a pre-computed dummy hash); wrong credentials → 401 "Invalid email or password"; status other than ACTIVE → 403; if 2FA is enabled, returns <code>twoFactorRequired</code> instead of tokens; otherwise <code>issueTokenPair</code> .
<code>issueTokenPair(user, ctx)</code>	Creates a <code>sessionId</code> (cuid2), signs the token pair and inserts a row into <code>user_sessions</code> (refresh hash, expiry from <code>exp</code> , device, IP).
<code>signTokenPair(user, sessionId)</code>	Signs access and refresh in parallel (<code>Promise.all</code>) — each with its own secret and lifetime.
<code>refresh(token, ctx)</code>	4 checks: (1) JWT signature and expiry; (2) the session exists, is not revoked and not expired; (3) the token hash matches the hash in the DB (<code>timingSafeEqual</code>); (4) the user is ACTIVE . If the signature is valid but the hash does not match, the token was already rotated (reuse): the session is revoked (<code>revokedAt = now</code>) and the response is 401. Otherwise it issues a new pair and overwrites the hash and expiry in the same session. Caveat: two tabs refreshing at the same moment with the same old token can log the user out (single-flight protects only within one tab).
<code>logout(userId, sessionId)</code>	Checks that the session belongs to the user; sets <code>revokedAt</code> .
<code>me(userId)</code>	Returns the user with <code>omit: { passwordHash, twoFactorSecret }</code> .
<code>twoFactorEnable(userId, purpose)</code>	Generates a code via <code>randomInt(100000, 1000000)</code> , hashes it with <code>bcrypt</code> , invalidates earlier live codes for the same purpose, creates a record (TTL 5 min), sends an email.
<code>verifyOtp(userId, purpose, otp)</code> (private)	Shared code check for both 2FA flows: takes the newest unused, unexpired code for the given purpose; 5 or more attempts → 400; wrong code → <code>attempts + 1</code> ; correct code → <code>verifiedAt = now</code> .
<code>twoFactorVerifyEnable</code>	Verifies an <code>ENABLE_TOGGLE</code> code and inverts <code>twoFactorEnabled</code> .
<code>twoFactorCreateOtpToken</code>	Sends a <code>LOGIN</code> code and signs the <code>twoFactorToken</code> .
<code>twoFactorVerifyLoginOtp</code>	Verifies the <code>twoFactorToken</code> , the user status and the <code>LOGIN</code> code, then <code>issueTokenPair</code> .
<code>listSessions / revokeSession / revokeOtherSessions</code>	Management of the device list (the user's own sessions only). <code>listSessions</code> returns only <code>revokedAt = null</code> and <code>expiresAt > now</code> , ordered by <code>lastActiveAt desc</code> .
<code>forgotPassword / forgotPasswordOtpVerification / resetPassword</code>	Three-step password recovery (section 7.8). <code>forgotPasswordOtpVerification</code> signs the reset token with <code>{ userId, passwordVersion }</code> ; <code>resetPassword</code> maps any <code>verifyAsync</code> failure to 400 "Session is Expired" and calls <code>UserService.resetForgottenPassword(userId, { password }, passwordVersion)</code> .

In `auth.service.ts` the methods are grouped under commented sections: authentication, token management, session management, token helpers, user profile, two-factor authentication, password recovery. The logic is unchanged by the regrouping.

5.6. How route protection works: Guard + Strategy + Decorator

```
// common/guards/jwt-auth.guard.ts
export class JwtAuthGuard extends AuthGuard('jwt') {}

// auth/strategies/jwt.strategy.ts
super({ jwtFromRequest: ExtractJwt.fromAuthHeaderAsBearerToken(),
  ignoreExpiration: false,
  secretOrKey: config.getOrThrow('JWT_ACCESS_SECRET') });

async validate(payload: JwtAccessPayload): Promise<AuthenticatedUser> {
  const session = await this.prisma.userSession.findUnique({
    where: { id: payload.sid }, include: { user: true } });
  if (!session || session.revokedAt || session.expiresAt <= new Date())
    throw new UnauthorizedException('Session is no longer valid');
  if (session.user.status !== UserStatus.ACTIVE)
    throw new UnauthorizedException('This account is not active');
  if (Date.now() - session.lastActiveAt.getTime() > LAST_ACTIVE_TOUCH_MS) // 60_000
    void this.prisma.userSession.updateMany({ // fire-and-forget
      where: { id: session.id, revokedAt: null },
      data: { lastActiveAt: new Date() } }).catch(() => undefined);
  return { userId: session.userId, email: session.user.email,
    userType: session.user.userType, sessionId: session.id };
}

// in the controller
@UseGuards(JwtAuthGuard)
@Get('me') me(@CurrentUser() user: AuthenticatedUser) {...}
```

- `@UseGuards(JwtAuthGuard)` → Passport invokes the `jwt` strategy. It extracts the token from the header and checks the **signature and expiry**.
- If the token is valid, `validate(payload)` is called. The strategy additionally goes **to the DB** for the session. This is the bridge between a "stateless JWT" and a "managed session".
- Nest puts the object returned from `validate` into `request.user`; the `@CurrentUser()` decorator extracts it into a method parameter.
- `email` and `userType` are taken **from the DB**, not from the token — so changes to them take effect immediately.
- `validate` also keeps `lastActiveAt` current: a fire-and-forget `updateMany` (errors swallowed), at most once per 60 s (`LAST_ACTIVE_TOUCH_MS`). Previously `lastActiveAt` changed only on refresh.

REMEMBER The price of instant logout

Every protected request makes a DB query for the session. This is a deliberate trade-off: a pure JWT needs no DB, but cannot "kick out" a user before the token expires. The project's choice is security and controllability.

5.7. UserService

Method	Essence
<code>findByEmail</code>	<code>findUnique</code> by email (citext → case-insensitive).
<code>update(userId, dto)</code>	Updates the profile without returning secret fields. An email uniqueness violation (Prisma <code>P2002</code>) → 409 "Email is already in use".
<code>changePassword</code>	Verifies <code>currentPassword</code> (otherwise 401), hashes the new one (bcrypt, cost 10) and in a single transaction updates the password + <code>passwordUpdatedAt</code> and revokes all other sessions.
<code>resetForgottenPassword(userId, { password }, expectedPasswordVersion)</code>	No old-password check. In one transaction: <code>user.updateMany({ where: { id, passwordUpdatedAt: new Date(expectedPasswordVersion) }, data: { passwordHash, passwordUpdatedAt: now } })</code> ; <code>count === 0</code> → 400 "Session is Expired" (replay or a concurrent second request) — so a reset token is single-use . Then it revokes all open sessions (the <code>currentSessionId</code> parameter is gone) and returns <code>{ user, message, success }</code> .

5.8. DTOs and input validation

DTO	Rules
LoginDto	email — @IsEmail; password — a non-empty string (login-dto.ts holds only this class)
RefreshTokenDto	refreshToken — @IsJWT (must look like a JWT)
TwoFactorDto	otp — a non-empty string (two-factor-dto.ts holds only this class)
TwoFactorVerifyLoginOtpDto	otp, twoFactorToken — non-empty strings (own file two-factor-verify-login-otp-dto.ts)
ForgotPasswordDto (new file)	only email — @IsEmail (step 1)
VerifyForgotPasswordOtpDto (new file)	email — @IsEmail; otp — a required non-empty string (step 2: POST /auth/forgot-password/verify-otp)
ResetPasswordDto	resetToken — @IsString @IsNotEmpty; password — @IsString @MinLength(8) @MaxLength(72) (same as ChangePasswordDto; has a spec)
ChangePasswordDto	currentPassword — non-empty; password — 8 to 72 characters
UpdateUserDto	all fields optional: fullName ≤ 120, email (@IsEmail), phone ≤ 32, profileImage

The 72 limit for passwords is not arbitrary: bcrypt only considers the first 72 bytes of a password. The `dto/index.ts` barrel re-exports all auth DTOs; the unused `create-auth.dto.ts` and `update-auth.dto.ts` were deleted.

5.9. Other pieces: User-Agent, email, error filter

- `@ParseUserAgent()` (`common/decorators/user-agent.decorator.ts`) — parses the `User-Agent` header with the `ua-parser-js` library; the result (`os`, `browser`, `device`) is stored in the session. `@Ip()` provides the IP.
- `EmailService` — sends emails through Resend: `sendOtp2FAuser` and `sendOtpForgotPassword`. If sending fails — 500 "Failed to send OTP email". The user's name in both HTML emails goes through `escapeHtml(userName)`, so user-controlled text is not inserted raw.
- `AllExceptionsFilter` — a single error format; 4xx are logged as warn, 5xx as error with the stack trace.

6. Frontend by layers (platform-admin)

6.1. Frontend file map

File	Role
<code>src/lib/storage.ts</code>	Wrapper over <code>localStorage</code> (<code>ACCESS_TOKEN</code> , <code>REFRESH_TOKEN</code> , <code>USER_TYPE</code>) with <code>try/catch</code>
<code>src/lib/axios.ts</code>	Two axios clients, interceptors, automatic refresh
<code>src/api/auth.ts</code>	The <code>authApi</code> object: one function per endpoint
<code>src/store/auth/authSlice.ts</code>	Redux: state and reducers only (<code>extraReducers</code> import the thunks)
<code>src/store/auth/thunks/*</code>	<code>session</code> , <code>profile</code> , <code>twoFactor</code> , <code>sessions</code> , <code>forgotPassword</code> <code>.thunks.ts</code> , <code>barrel index.ts</code> , <code>thunk-test-utils.ts</code>
<code>src/store/auth/auth.helpers.ts</code>	<code>buildAvatarName</code>
<code>src/lib/validators.ts</code>	<code>validateEmail</code> , <code>validatePassword</code> (shared form validators)
<code>src/store/auth/auth.types.ts</code>	The <code>AuthState</code> type
<code>src/types/auth.ts</code> , <code>types/user.ts</code>	Request/response types, type guards <code>isCompleteLoginResponse</code> , <code>isTwoFactorRequiredResponse</code>
<code>src/routes/index.tsx</code> , <code>routes/route-modules.ts</code>	Route table (<code>/auth/*</code> public, <code>/dashboard/*</code> protected); <code>route-modules.ts</code> maps lazy imports (<code>routeModules.login</code> → <code>@/pages/auth/login/LoginPage</code> , <code>forgotPassword</code> → <code>@/pages/auth/forgot-pasword</code>)
<code>src/components/ProtectedRoute.tsx</code>	Blocks access without a token, loads the profile
<code>src/layouts/AuthLayout.tsx</code>	Wrapper for the sign-in screens; if a token exists, redirects to <code>/dashboard</code>
<code>src/pages/auth/login/</code>	<code>LoginPage.tsx</code> , <code>validateLogin.ts</code> , <code>hooks/</code> (<code>useLoginCredentials</code> , <code>useLoginValidation</code> , <code>useLoginSubmit</code> , <code>useRedirectTarget</code> , <code>barrel index.ts</code>)
<code>src/pages/auth/forgot-pasword/</code>	(sic — typo in the folder name) <code>ForgotPasswordPage.tsx</code> , <code>hooks/</code> (<code>useForgotPasswordEmail</code> , <code>useForgotPasswordValidation</code> , <code>useForgotPasswordSubmit</code>)
<code>src/pages/auth/*.tsx</code>	<code>Verify2FaOtpPage</code> , <code>VerifyForgotOtpPage</code> , <code>ResetPasswordPage</code>
<code>src/pages/dashboard/commission-pay outs/</code>	renamed from <code>commision-payouts</code> (route path <code>commission-payouts</code> , <code>routeModules.commissionPayouts</code>)
<code>src/pages/dashboard/account/*</code>	Profile, password change, 2FA, session list
<code>src/components/AppSidebar.tsx</code>	Logout button (<code>fetchLogout</code>)
<code>src/lib/redirect.ts</code>	<code>isSafeRedirectPath</code> — protection against open redirect
<code>src/hooks/useOtpCountdown.ts</code>	5-minute timer for the password reset code
<code>*.test.ts(x)</code> next to the sources	Vitest + Testing Library: thunks, hooks, pages, validators, axios, routes (the repo now has 37 spec/test files in total, backend included)

6.2. Token storage: `lib/storage.ts`

- Tokens are kept in **localStorage** under the keys `ACCESS_TOKEN` and `REFRESH_TOKEN`; `USER_TYPE` is stored there too.
- Every operation is wrapped in `try/catch` (`safeGetItem`, `safeSetItem...`): if storage is unavailable (private mode, cookies blocked), the application does not crash — the login simply will not survive a reload.

- At startup, `initialState` in `authSlice` reads the tokens from storage — which is why the user stays "signed in" after F5.

6.3. `lib/axios.ts`: the heart of frontend authentication

```
export const Axios = axios.create({ baseURL }); // main client
const refreshClient = axios.create({ baseURL }); // separate one – only for /auth/refresh

const PUBLIC_AUTH_PATHS = ["/auth/login", "/auth/refresh"];

Axios.interceptors.request.use((config) => {
  if (isPublicAuthPath(config.url)) return config; // login has no token yet
  const t = storage.getAccessToken();
  if (t) config.headers.Authorization = `Bearer ${t}`;
  return config;
});

let pendingRefresh: Promise<string> | null = null;
function fetchRefreshToken() { // "single-flight"
  pendingRefresh ??= requestTokenPair().finally(() => { pendingRefresh = null; });
  return pendingRefresh;
}

Axios.interceptors.response.use(r => r, async (error) => {
  const req = error.config;
  const hadAuthHeader = Boolean(req?.headers?.Authorization);
  if (error.response?.status !== 401 || !req || req._retry || !hadAuthHeader)
    return Promise.reject(error);
  req._retry = true;
  try { token = await fetchRefreshToken(); }
  catch (e) { storage.clearStorage(); window.location.href = "/auth/login"; throw e; }
  req.headers.Authorization = `Bearer ${token}`;
  return Axios(req); // retry the original request
});
```

Technique	Why
A separate <code>refreshClient</code>	The refresh request does not pass through the main client's interceptors — otherwise its own 401 would trigger refresh again (an infinite loop).
<code>_retry = true</code>	A single request is retried at most once.
<code>hadAuthHeader</code>	A 401 is handled only for requests that carried a token. A wrong password at login also yields 401, but refreshing the token makes no sense there.
<code>pendingRefresh</code> (single-flight)	If 5 requests get a 401 simultaneously, only one refresh goes to the server and the rest wait on the same Promise. This is critical: the refresh token is single-use, so a second parallel refresh with the same token would fail.
<code>window.location.href = '/auth/login'</code>	If the refresh failed — a full storage reset (<code>storage.clearStorage()</code> , formerly the misspelled <code>clearStoradge</code> ; it clears all of <code>localStorage</code>) and a hard redirect to the sign-in page.

6.4. `Redux`: `authSlice`

- State** (`AuthState`): `user`, `accessToken`, `refreshToken`, `userType`, `error`, `twoFactorToken`, `twoFactorRequired`, `sessions` and a set of loading flags (`isLoginLoading`, `isMeLoading`, `isLogoutLoading`, `isForgotPasswordLoading`, ...).
- The slice holds only state and reducers** (`clearAuthError` plus `extraReducers`); the thunks live in `store/auth/thunks/` and pages import them from `@/store/auth/thunks`. Only `fetchForgotPassword` has pending/fulfilled/rejected reducers (they set `isForgotPasswordLoading`); the verify and reset thunks have no reducers.
- Thunks**, by file: `session` — `fetchLogin`, `fetchMe`, `fetchLogout`, `verify2FaLoginOtp`; `profile` — `updateUser`, `changePassword`; `twoFactor` — `twoFactorGenerateOtp`, `verifyEnableDisableTwoFactor`; `sessions` — `fetchSessionsList`, `revokeSession`, `revokeOtherSessions`; `forgotPassword` — `fetchForgotPassword`, `fetchForgotPasswordOtpVerify`, `resetForgottenPassword`.

- A uniform pattern: `try { await authApi.x() } catch { return rejectWithValue(getAxiosErrorMessage(...)) }`. A page calls `dispatch(thunk).unwrap()` inside `try/catch` and shows the error text.
- `getAxiosErrorMessage (lib/api.error.ts)` extracts `message` from the backend response (a string or an array — in which case the first element is taken), otherwise substitutes a fallback text.
- Tokens are saved to `localStorage` **inside the thunk** (`storage.setTokens`) and reach Redux through the reducer.
- `fetchMe.rejected` with `unauthorized: true` (a real 401) clears the tokens; a network error or 5xx does **not** log the user out.

Type guards. The `/auth/login` response comes in two forms: "tokens" or "2FA required".

`isCompleteLoginResponse` and `isTwoFactorRequiredResponse` (`types/auth.ts`) check the shape of the response at runtime and at the same time narrow the type in TypeScript.

6.5. Routes and page protection

```
<Route path="auth" element={<AuthLayout />}>    // public
  login | 2fa | forgot-password | forgot-password/otp | reset-password
</Route>
<Route element={<ProtectedRoute />}>            // protected
  <Route path="dashboard" element={<DashboardLayout />}> ... my-account ...
</Route>
```

Component	Logic
ProtectedRoute	No <code>accessToken</code> in Redux → <code><Navigate to="/auth/login?redirect_uri=<current path>"></code> . Token present → <code>dispatch(fetchMe())</code> ; while <code>user === null</code> shows "Loading..."; then <code><Outlet /></code> .
AuthLayout	If an <code>accessToken</code> already exists, calls <code>navigate('/dashboard')</code> — a signed-in user does not need the sign-in screen.
RootRedirect	The root <code>/</code> always leads to <code>/auth/login</code> .
route-modules.ts	Lazy-import map used for routes and prefetch (<code>PrefetchLink</code>): <code>login</code> → <code>@/pages/auth/login/LoginPage</code> , <code>forgotPassword</code> → <code>@/pages/auth/forgot-pasword</code> , <code>commissionPayouts</code> → <code>.../commission-payouts/CommissionPayoutsPage</code> .
isSafeRedirectPath	Allows only paths of the form <code>/something</code> : must start with <code>/</code> , but not with <code>//</code> and not with <code>/\</code> . This way <code>?redirect_uri=//evil.com</code> will not turn into a jump to a foreign site.

6.6. The authentication screens and how they communicate

Screen	What it does
LoginPage (login/)	A thin page over hooks; the form is <code>noValidate</code> , per-field errors appear on submit and clear on edit (<code>validateLogin</code> → <code>lib/validators.ts</code> : email required + simple regex, password required only — length is deliberately not enforced at login). <code>useLoginSubmit</code> dispatches <code>fetchLogin(...).unwrap()</code> , shows <code>toast.success/error</code> , then a full response → <code>navigate(redirectTo)</code> (<code>?redirect_uri</code> checked by <code>isSafeRedirectPath</code> , default <code>/dashboard</code>); 2FA → <code>navigate('/auth/2fa', { state: { twoFactorToken, email, redirectTo } })</code> .
Verify2FaOtpPage	Reads <code>twoFactorToken</code> , <code>email</code> , <code>redirectTo</code> from <code>location.state</code> ; no token → back to <code>/auth/login</code> ; 6-digit <code>InputOTP</code> ; <code>verify2FaLoginOtp</code> ; the loading flag is <code>isVerify2FaLoginOtpLoading</code> (it used to read the wrong one).
ForgotPasswordPage (forgot-password/)	Hooks <code>useForgotPasswordEmail</code> , <code>useForgotPasswordValidation</code> (<code>validateEmail</code>), <code>useForgotPasswordSubmit</code> : loading comes from <code>state.auth.isForgotPasswordLoading</code> ; <code>fetchForgotPassword</code> → <code>navigate('/auth/forgot-password/otp', { state: { email, createdAt } })</code> .
VerifyForgotOtpPage	Timer <code>useOtpCountdown(createdAt, 5)</code> ; after it expires — a "Resend Otp" button; <code>fetchForgotPasswordOtpVerify</code> → <code>navigate('/auth/reset-password', { state: { email, resetToken }, replace: true })</code> .
ResetPasswordPage	Client-side check that the passwords match; without <code>email/resetToken</code> in state — back to <code>/auth/forgot-password</code> ; both password inputs have <code>minLength={8}</code> <code>maxLength={72}</code> (matching the backend DTO); <code>resetForgottenPassword</code> ; a "Password Updated" screen.
ProfileDetailsPage + components	<code>EditProfile</code> , <code>ChangePassword</code> , <code>TwoFactorAuthentication</code> (dialogs), <code>SessionsList</code> .

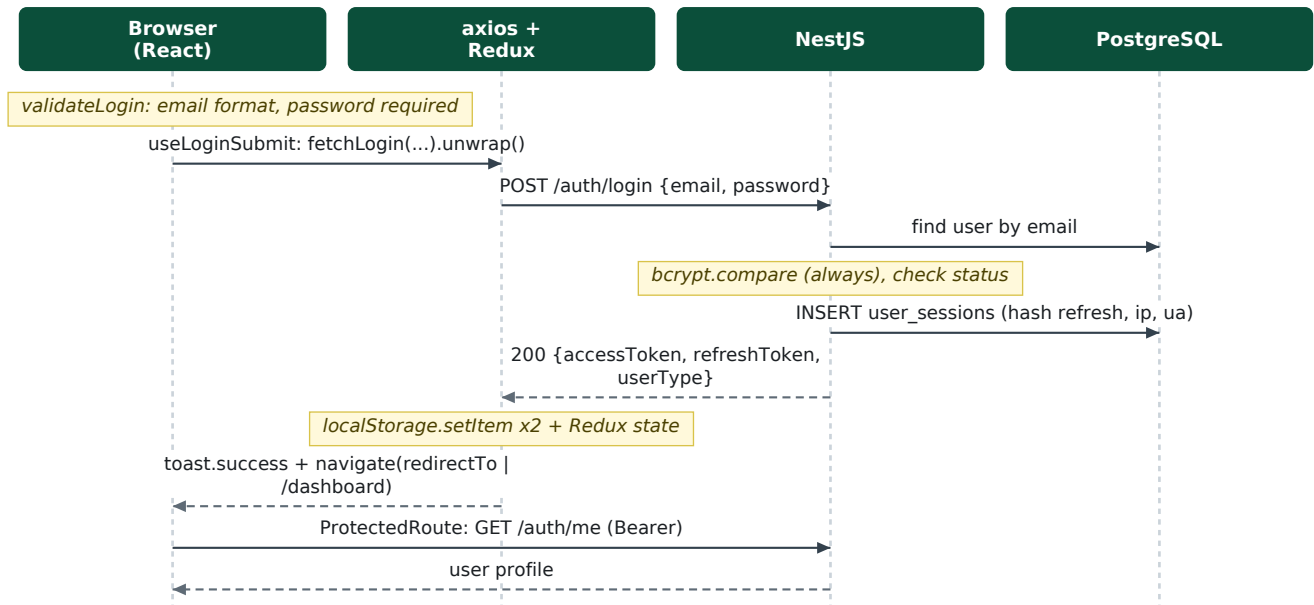
REMEMBER Intermediate tokens live in `location.state`

`twoFactorToken` and `resetToken` are passed between screens through router state (`navigate(..., { state })`), not through Redux or localStorage. They do not end up in the URL (they are not exposed in the address bar or in shared links). Opening `/auth/2fa` or `/auth/reset-password` directly by link is not possible: without `state` the page immediately sends the user back.

7. Step-by-step flows

This is the main part of the notes. For each flow: a diagram, then numbered steps stating **where** each action happens.

7.1. Login without 2FA



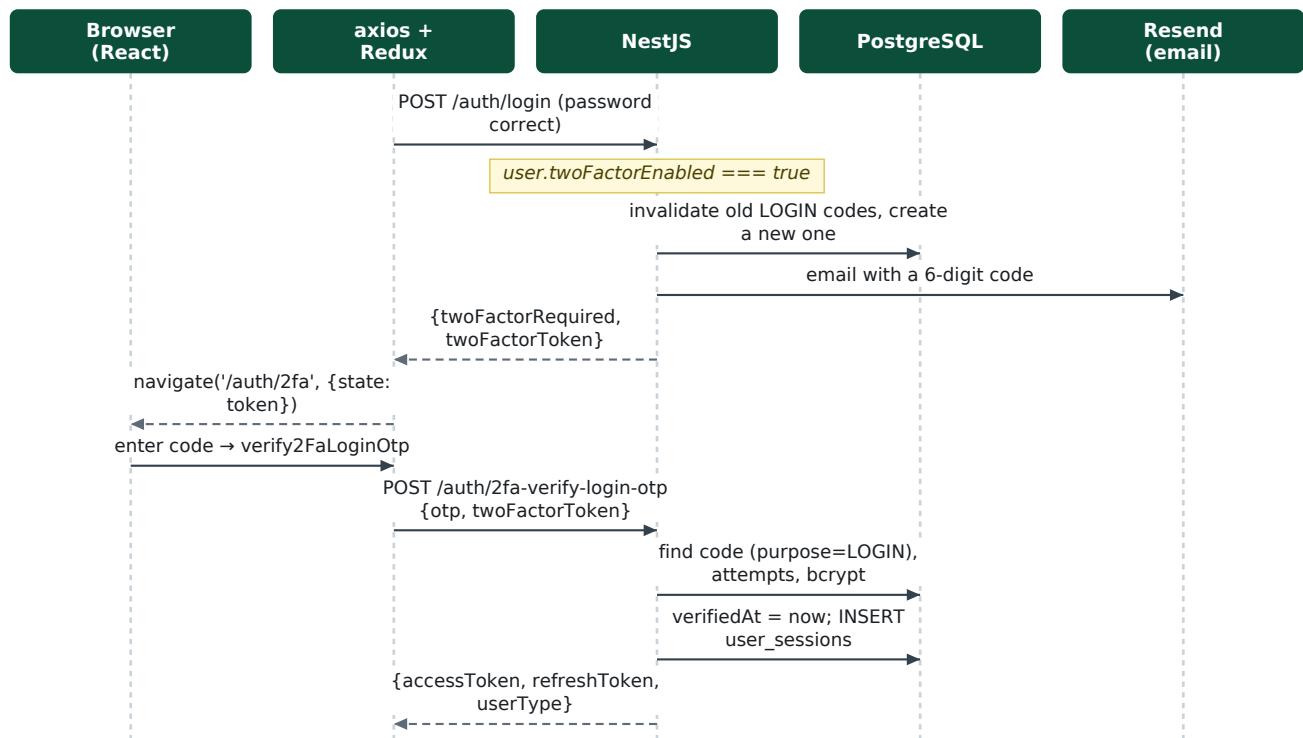
#	Where	What happens
1	Frontend	The user enters an email and password on <code>pages/auth/login/LoginPage.tsx</code> , a thin page; its logic lives in the hooks <code>useLoginCredentials</code> (field state), <code>useLoginValidation</code> and <code>useRedirectTarget</code> . The form is <code>noValidate</code> : checks are done by <code>validateLogin</code> on submit (<code>validateEmail</code> : required + a simple regex; <code>validatePassword</code> : required only , the login form deliberately does not enforce length). Errors are shown per field and cleared on edit. This is UX only, not security.
2	Frontend	<code>useLoginSubmit</code> : if validation passes → <code>dispatch(fetchLogin(credentials)).unwrap()</code> (the thunk is in <code>store/auth/thunks/session.thunks.ts</code>) → <code>authApi.login</code> → <code>POST /auth/login</code> . The interceptor recognizes a public path and does not add <code>Authorization</code> .
3	Backend	Rate limit of 10 requests per minute; <code>ValidationPipe</code> validates <code>LoginDto</code> (email format, non-empty password).
4	Backend	<code>AuthController.login</code> collects the context: IP (<code>@Ip</code>) and the parsed <code>User-Agent</code> (<code>@ParseUserAgent</code>) → <code>{ ipAddress, device, os, browser }</code> .
5	Backend	<code>AuthService.login</code> : <code>findByEmail</code> ; then <code>bcrypt.compare(password, user?.passwordHash ?? ABSENT_USER_PASSWORD_HASH)</code> runs always , so the response time does not reveal whether the email exists.
6	Backend	No such user or wrong password → 401 "Invalid email or password" (identical in both cases). Status other than <code>ACTIVE</code> → 403 "This account is not active".
7	Backend	2FA disabled → <code>issueTokenPair</code> : a <code>sessionId</code> is created, access (15 min) and refresh (30 days) tokens are signed, the HMAC hash of the refresh token is written to <code>user_sessions</code> , and <code>expiresAt</code> is taken from its <code>exp</code> .
8	Backend	Response 200: <code>{ accessToken, refreshToken, userType }</code> .
9	Frontend	The thunk checks the response shape (<code>isCompleteLoginResponse</code>), then calls <code>storage.setTokens</code> and <code>storage.setUserType</code> . The <code>fetchLogin.fulfilled</code> reducer puts the tokens into Redux.
10	Frontend	<code>useLoginSubmit</code> shows <code>toast.success("Signed in successfully")</code> and calls <code>navigate(redirectTo, { replace: true })</code> . <code>redirectTo</code> comes from <code>useRedirectTarget</code> : the <code>?redirect_uri</code> parameter (if it passed <code>isSafeRedirectPath</code>), otherwise <code>/dashboard</code> .

#	Where	What happens
11	Frontend	<code>ProtectedRoute</code> sees a token → <code>fetchMe()</code> → GET <code>/auth/me</code> with Bearer. Until the profile arrives it shows "Loading...", then <code>DashboardLayout</code> is rendered.

WARNING Login failure

The thunk's `catch` calls `storage.clearStorage()` (which still wipes all of `localStorage`) and `rejectWithValue(message)`. `useLoginSubmit` catches the rejection, shows `toast.error(message)` and the same text under the form.

7.2. Login with two-factor authentication



#	Where	What happens
1	Backend	The password is correct but <code>twoFactorEnabled = true</code> → <code>twoFactorCreateOtpToken</code> : calls <code>twoFactorEnable(userId, LOGIN)</code> . A code is generated with <code>randomInt(100000, 1000000)</code> , its bcrypt hash is written to <code>two_factor_otps</code> with a 5-minute TTL, earlier live codes for the same purpose are invalidated, and the email is sent through Resend.
2	Backend	A <code>twoFactorToken</code> is signed (JWT_2FA_SECRET, 15 min, payload { <code>userId</code> }). There are no access tokens yet. Response: { <code>twoFactorRequired</code> : true, <code>twoFactorToken</code> , <code>message</code> }.
3	Frontend	In <code>useLoginSubmit</code> , <code>isTwoFactorRequiredResponse</code> → navigation to <code>/auth/2fa</code> with <code>state</code> : { <code>twoFactorToken</code> , <code>email</code> , <code>redirectTo</code> }.
4	Frontend	<code>Verify2FaOtpPage</code> : the user enters 6 digits → POST <code>/auth/2fa-verify-login-otp</code> (limit 5 per minute). The page now reads the correct flag <code>isVerify2FaLoginOtpLoading</code> , so the button and input are disabled while the request is in flight.
5	Backend	<code>verifyAsync(twoFactorToken)</code> : failure → 400 "Session is Expired". Then: the user exists and is ACTIVE. Note: the <code>twoFactorToken</code> itself is not single-use; it is valid for 15 minutes, but is useless without a correct code.
6	Backend	<code>verifyOtp(userId, LOGIN, otp)</code> : the newest unused, unexpired code → otherwise 400 "Otp expired or invalid"; <code>attempts ≥ 5</code> → 400 "Too many otp attempts"; wrong code → <code>attempts + 1</code> and 400 "Invalid Otp"; correct → <code>verifiedAt = now</code> .
7	Backend	<code>issueTokenPair</code> : from here it is the same as in 7.1, a session is created and the tokens are returned.

#	Where	What happens
8	Frontend	Tokens are saved to localStorage and Redux, then <code>navigate(redirectTo)</code> .

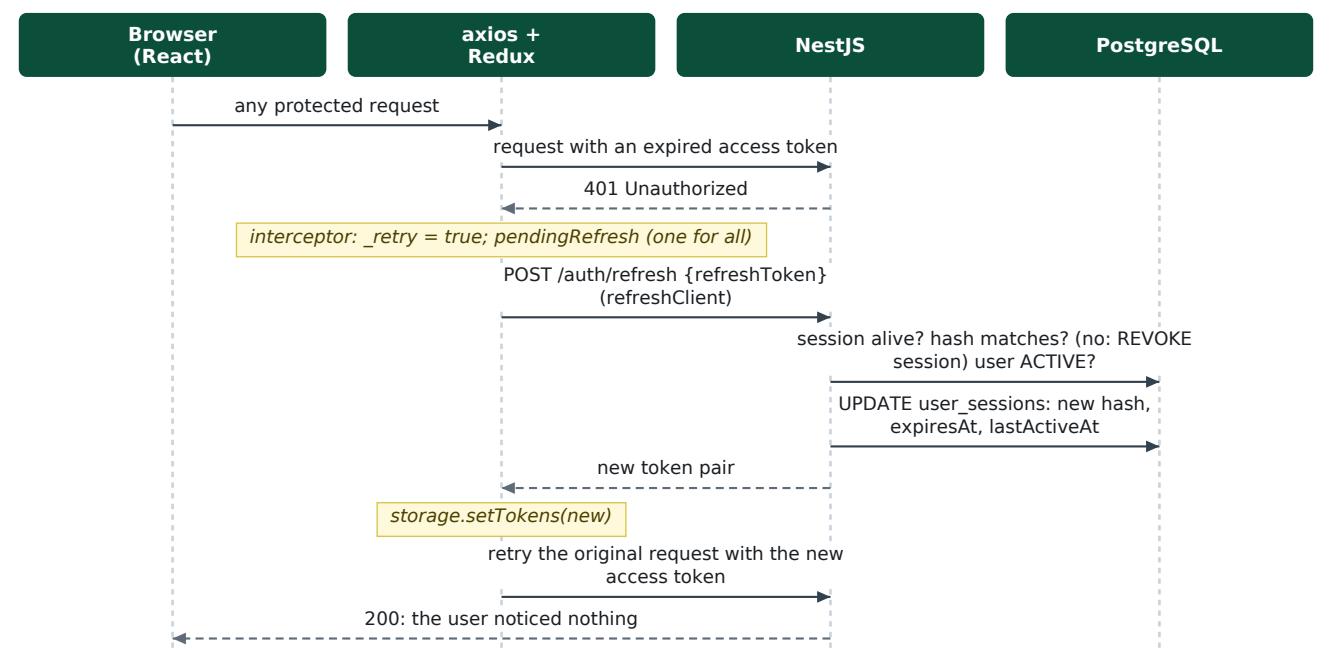
REMEMBER Why both `twoFactorToken` and a code

These are two different proofs. The `twoFactorToken` (signed by the server) proves that "the password was already verified", and the server does not store that state. The **emailed code** proves control of the mailbox. Without the token, codes cannot be tried against someone else's account; without the code, the token is useless.

7.3. Request to a protected endpoint

#	Where	What happens
1	Frontend	The request interceptor takes <code>ACCESS_TOKEN</code> from localStorage and adds <code>Authorization: Bearer <token></code> .
2	Backend	<code>ThrottlerGuard</code> (rate limit) → <code>JwtAuthGuard</code> → Passport strategy <code>jwt</code> : extracts the Bearer token, verifies the signature (<code>JWT_ACCESS_SECRET</code>) and the expiry (<code>ignoreExpiration: false</code>). Failure → 401.
3	Database	<code>JwtStrategy.validate(payload)</code> : looks up <code>user_sessions</code> by <code>payload.sid</code> together with the user.
4	Backend	No session, <code>revokedAt</code> set, or <code>expiresAt</code> in the past → 401 "Session is no longer valid". User not <code>ACTIVE</code> → 401 "This account is not active".
5	Backend	Activity tracking: if <code>lastActiveAt</code> is older than 60 s (<code>LAST_ACTIVE_TOUCH_MS</code>), <code>validate</code> fires an <code>updateMany</code> on the session (fire-and-forget: it does not block the request, errors are swallowed). So <code>lastActiveAt</code> follows real activity, not just refreshes.
6	Backend	<code>{ userId, email, userType, sessionId }</code> is placed in <code>request.user</code> ; <code>@CurrentUser()</code> hands it to the controller.
7	Backend	<code>ValidationPipe</code> , the controller, the service and Prisma run; the response goes to the client.

7.4. Automatic token refresh



#	Where	What happens
1	Frontend	A request got 401. The interceptor checks the conditions: status 401; the request has not been retried yet (<code>_retry</code>); the request carried an <code>Authorization</code> header (i.e. it is not a login).

#	Where	What happens
2	Frontend	<code>fetchRefreshToken()</code> : if a refresh is already in flight, all requests wait on the same Promise (<code>pendingRefresh</code>). Otherwise <code>requestTokenPair()</code> takes <code>REFRESH_TOKEN</code> and sends <code>POST /auth/refresh</code> through <code>refreshClient</code> .
3	Backend	<code>AuthService.refresh</code> , step 1: <code>verifyAsync</code> with <code>JWT_REFRESH_SECRET</code> (signature and expiry) → otherwise 401 "Invalid refresh token".
4	Backend	Step 2: the session <code>payload.sessionId</code> exists, is not revoked and has not expired.
5	Backend	Step 3: the HMAC of the presented token is compared with <code>refreshTokenHash</code> via <code>timingSafeEqual</code> . Mismatch means a validly signed but already rotated token is being replayed (reuse detection): the session is revoked (<code>updateMany revokedAt = now</code>) and only then 401 is returned. The legitimate owner has to sign in again. Caveat: two tabs refreshing at the same moment with the same old token can log the user out (single-flight protects only within one tab).
6	Backend	The user must be ACTIVE (otherwise 403).
7	Backend	Step 4 is rotation : <code>signTokenPair</code> creates a new pair, and in the same session <code>refreshTokenHash</code> , <code>expiresAt</code> , <code>lastActiveAt</code> , IP, OS and browser are updated.
8	Frontend	<code>storage.setTokens(new)</code> , and the original request is retried with the new access token. If the refresh fails, <code>storage.clearStorage()</code> and <code>window.location.href = '/auth/login'</code> .

REMEMBER The session is "sliding"

Every successful refresh extends `expiresAt` by another refresh-token lifetime (30 days). If a user signs in regularly, the session never expires; if the user is absent for 30 days, a new login is required.

7.5. Logout and session management

#	Where	What happens
1	Frontend	The logout button in <code>AppSidebar</code> → <code>dispatch(fetchLogout())</code> → <code>POST /auth/logout</code> (Bearer).
2	Backend	<code>AuthService.logout(userId, sessionId)</code> : the session must belong to the user (otherwise 400), then <code>updateMany({ id, revokedAt: null }, { revokedAt: now })</code> .
3	Backend	From this moment <code>JwtStrategy</code> rejects the access token of this session (which could still have up to 15 minutes to live), and the refresh token fails the session check.
4	Frontend	In both <code>fulfilled</code> and <code>rejected</code> , the reducer clears Redux and calls <code>storage.clearStorage()</code> , so the logout happens locally even if the server is unreachable.
5	Frontend	Device list: <code>SessionsList</code> on mount → <code>GET /auth/sessions</code> → the server returns only sessions with <code>revokedAt = null</code> and <code>expiresAt > now</code> (expired ones are hidden), ordered by <code>lastActiveAt</code> descending, with an <code>isCurrent</code> flag (the session from the token).
6	Backend	<code>POST /auth/sessions/:id/revoke</code> revokes one session (owner check: someone else's session cannot be revoked). <code>POST /auth/sessions/revoke-others</code> revokes all except the current one (NOT: <code>{ id: currentSessionId }</code>).
7	Frontend	After a revoke the UI requests the session list again.

7.6. Enabling and disabling 2FA

#	Where	What happens
1	Frontend	<code>TwoFactorAuthentication</code> (dialog): the "Send OTP" button → <code>POST /auth/2fa-generate-otp</code> (Bearer, limit 5/min).
2	Backend	<code>twoFactorEnable(userId)</code> with purpose <code>ENABLE_TOGGLE</code> : code, bcrypt hash, 5-minute TTL, invalidation of earlier codes for this purpose, email.
3	Frontend	The dialog switches to a 6-digit input → <code>POST /auth/2fa-verify-enable { otp }</code> (limit 5/min).
4	Backend	<code>verifyOtp(ENABLE_TOGGLE)</code> , then <code>twoFactorEnabled: !user.twoFactorEnabled</code> ; this is a toggle, not an "enable". Response <code>{ success, message: 'Two-factor enabled disabled', data: { twoFactorEnabled } }</code> .
5	Frontend	The reducer updates <code>state.user.twoFactorEnabled</code> , and the Enabled/Disabled badge changes.

REMEMBER Why the `purpose` field exists

The login code (`LOGIN`) and the 2FA-toggle code (`ENABLE_TOGGLE`) are stored in one table but differ by purpose. `verifyOtp` looks for a code **of its own purpose only**, so the code from a "confirm sign-in" email cannot be used to disable 2FA.

7.7. Changing the password (current one is known)

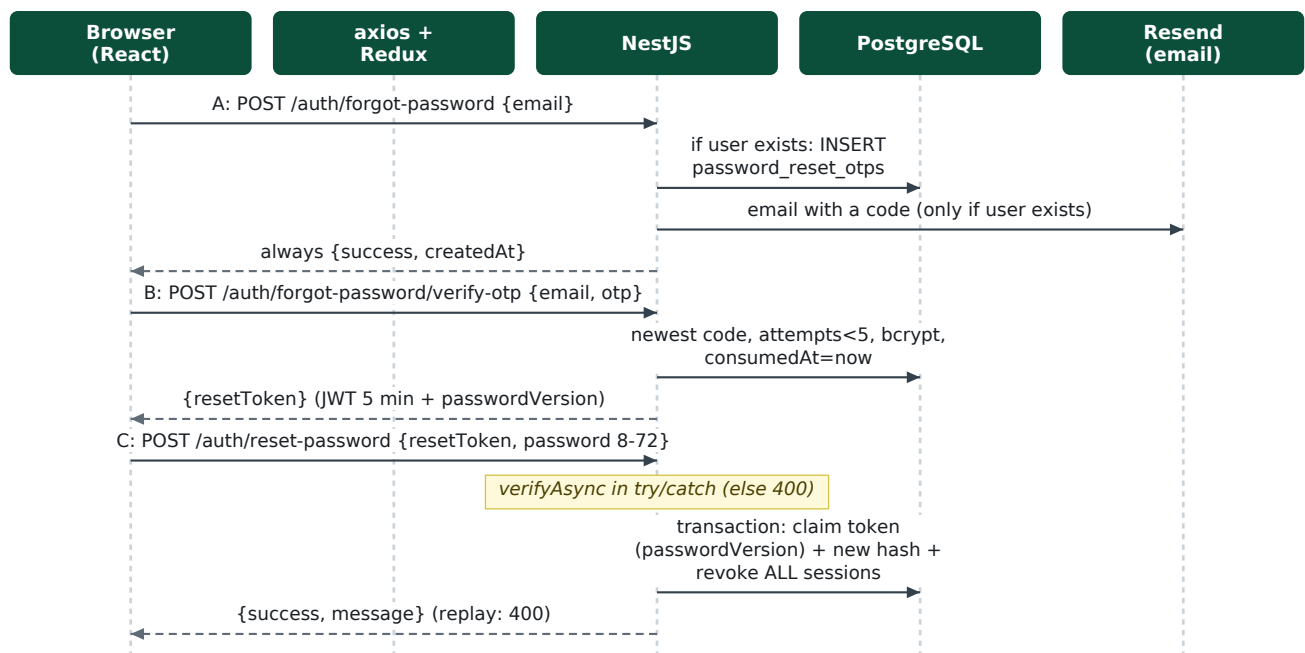
#	Where	What happens
1	Frontend	<code>ChangePassword</code> dialog: current / new / confirmation fields. A mismatch between the new passwords is caught on the client. The button is disabled until everything is filled in.
2	Frontend	<code>PUT /users/change-password { currentPassword, password }</code> (Bearer).
3	Backend	<code>ChangePasswordDto</code> : the new password is 8-72 characters.
4	Backend	<code>UserService.changePassword</code> : <code>bcrypt.compare(currentPassword, hash)</code> → wrong → 401 "Current password is incorrect".
5	Database	In one transaction : the new <code>passwordHash</code> + <code>passwordUpdatedAt</code> , and <code>revokedAt = now</code> for all other sessions of the user (except the current one, <code>id: { not: currentSessionId }</code>).
6	Frontend	The response updates <code>state.user</code> ; on the account page the "changed ... ago" label changes.

REMEMBER Changing the password signs out the other devices

If a password was stolen and the attacker is already signed in, changing the password immediately revokes that session: the attacker's access token stops passing `JwtStrategy`, and the refresh token fails the session check.

7.8. Password recovery (forgotten password)

Three steps, three "proofs": the **emailed code** (control of the mailbox) → the `resetToken` (code passed) → the **new password**.



#	Where	What happens
1	Frontend	<code>pages/auth/forgot-password/ForgotPasswordPage.tsx</code> (sic, the folder name has a typo) with the hooks <code>useForgotPasswordEmail</code> , <code>useForgotPasswordValidation</code> (<code>validateEmail</code>) and <code>useForgotPasswordSubmit</code> . After validation the hook dispatches <code>fetchForgotPassword({ email })</code> (thunk in <code>thunks/forgotPassword.thunks.ts</code>); the loading state comes from the new flag <code>state.auth.isForgotPasswordLoading</code> ; success and error are shown as toasts.
2	Backend	<code>forgotPassword(email)</code> : if the user exists, a 6-digit code, bcrypt hash, 5-minute TTL, email. The response is always <code>{ success: true, message, data: { createdAt } }</code> (for a non-existent email <code>createdAt</code> is simply "now"). This way it is impossible to learn which emails are registered.
3	Frontend	<code>useForgotPasswordSubmit</code> navigates to <code>/auth/forgot-password/otp</code> with <code>state: { email, createdAt }</code> . The <code>useOtpCountdown</code> hook counts down the remaining 5 minutes from <code>createdAt</code> ; when it runs out, "Resend Otp" is shown.
4	Backend	<code>forgotPasswordOtpVerification(email, otp)</code> : the user (otherwise 400 "Otp does not match"); the newest unused, unexpired code; <code>attempts ≥ 5 → 400</code> ; wrong <code>→ attempts + 1</code> ; correct <code>→ consumedAt = now</code> (the code is single-use). The request body is <code>VerifyForgotPasswordOtpDto</code> (<code>email + required otp</code>). Note: a new forgot-password request does not invalidate older unused codes.
5	Backend	A <code>resetToken</code> is signed (<code>JWT_RESET_SECRET</code> , 5 minutes) with the payload <code>PasswordResetTokenPayload { userId, passwordVersion }</code> , where <code>passwordVersion</code> is the user's <code>password_updated_at</code> in epoch ms at the time of issue. The token is returned to the client.
6	Frontend	Navigation to <code>/auth/reset-password</code> (<code>replace: true</code>) with <code>state: { email, resetToken }</code> .
7	Frontend	<code>ResetPasswordPage</code> : the inputs have <code>minLength={8}</code> <code>maxLength={72}</code> ; check that the two passwords match <code>→ POST /auth/reset-password { resetToken, password }</code> .
8	Backend	<code>ResetPasswordDto</code> : <code>resetToken</code> is a non-empty string, <code>password</code> is 8-72 characters (72 is the bcrypt limit; the same rule as <code>ChangePasswordDto</code>). <code>resetPassword</code> wraps <code>verifyAsync</code> in <code>try/catch</code> : an expired or malformed token <code>→ 400 "Session is Expired"</code> (no more 500).
9	Backend	<code>UserService.resetForgottenPassword(userId, { password }, payload.passwordVersion)</code> : in one transaction <code>user.updateMany({ where: { id, passwordUpdatedAt: new Date(passwordVersion) }, data: { passwordHash, passwordUpdatedAt: now } })</code> . <code>count = 0</code> (replay or a concurrent second request) <code>→ 400 "Session is Expired"</code> . Then all open sessions of the user are revoked (there is no current session to spare). Response <code>{ user, message, success }</code> .
10	Frontend	The "Password Updated" screen and a "Back to sign in" button. Tokens are not issued ; after a reset the user must sign in again.

REMEMBER Why the reset token is single-use

Changing the password changes `password_updated_at`, so the `passwordVersion` inside any already issued reset token no longer matches. A second use of the same token (a replay, or two parallel requests) updates zero rows and gets 400. The `updateMany` with a condition is an atomic "claim": no extra table or `jti` is needed.

7.9. Editing the profile

#	Where	What happens
1	Frontend	<code>EditProfile</code> (dialog): a form with <code>fullName</code> , <code>email</code> , <code>phone</code> , <code>profileImage</code> ; the button is enabled only if something changed.
2	Frontend	<code>PUT /users/me</code> (Bearer) with these fields.
3	Backend	<code>UpdateUserDto</code> + <code>whitelist</code> : extra fields such as <code>userType</code> or <code>status</code> are rejected.
4	Backend	<code>UserService.update</code> → Prisma <code>update</code> with <code>omit</code> . An email already taken → Prisma <code>P2002</code> → 409 "Email is already in use".
5	Frontend	The <code>updateUser.fulfilled</code> reducer (thunk in <code>thunks/profile.thunks.ts</code>) updates <code>state.user</code> and recomputes the avatar initials with <code>buildAvatarName</code> (<code>auth.helpers.ts</code>).

8. Strengths of the current design

These are the practices the team should understand and be able to explain; they are already in place and should be preserved when the code changes.

Practice	Where	Why it helps
Passwords hashed with bcrypt (cost 10)	<code>login</code> , <code>UserService</code> , <code>seed.ts</code>	A DB leak does not expose passwords
Dummy hash for a non-existent user	<code>ABSENT_USER_PASSWORD_HASH</code>	Response time does not reveal whether the email exists
Identical message "Invalid email or password"	<code>login</code>	No user enumeration
Identical <code>forgot-password</code> response for any email	<code>forgotPassword</code>	The same protection for password reset
Refresh stored in the DB as an HMAC, not the token itself	<code>hashRefreshToken</code>	A DB leak does not yield working tokens
Constant-time token comparison	<code>timingSafeEqual</code>	Protection against timing attacks
Refresh rotation on every renewal	<code>refresh</code>	A stolen old refresh token stops working
Refresh-token reuse detection: a stale token revokes the session	<code>refresh</code>	A stolen or leaked rotated token kills the session instead of silently failing
Session and status checked on every request	<code>JwtStrategy.validate</code>	Instant logout and blocking
Separate secrets for 4 kinds of tokens	<code>.env</code> , <code>signTokenPair</code>	A token of one kind cannot be passed off as another
OTP: <code>randomInt</code> , bcrypt hash, 5-min TTL, 5 attempts	<code>twoFactorEnable</code> , <code>verify0tp</code>	Codes cannot be guessed or reused
<code>purpose</code> field on 2FA codes	<code>two_factor_otps</code>	Codes from different flows do not mix

Practice	Where	Why it helps
Rate limiting on sensitive routes	<code>@Throttle</code>	Makes brute-forcing passwords and codes harder
Transaction on password change and session revocation	<code>\$transaction</code>	Password and sessions change atomically
Single-use reset token (<code>passwordVersion</code> + conditional <code>updateMany</code>)	<code>resetPassword</code> , <code>resetForgottenPassword</code>	Replay or a parallel second request fails with 400
Reset: <code>try/catch</code> on the token, 8-72 password validation, all sessions revoked	<code>resetPassword</code> , <code>ResetPasswordDto</code>	Clean 400 instead of 500; a stolen session cannot survive a reset
<code>lastActiveAt</code> updated at most once per 60 s, fire-and-forget	<code>JwtStrategy.validate</code>	Accurate device list without a DB write on every request
Expired sessions are hidden from the device list	<code>listSessions</code>	The list shows only sessions that can actually be used
User-controlled text is HTML-escaped in emails	<code>escapeHtml</code> in <code>email.service.ts</code>	No HTML injection through the user name
Shared validators and form hooks, covered by tests	<code>lib/validators.ts</code> , <code>pages/auth/login/*</code> , <code>*.spec.ts</code> , <code>*.test.tsx</code>	Logic is separated from the UI and checked (38 spec/test files: backend and frontend vitest tests, Testing Library on the frontend)
Thunks split from the slice	<code>store/auth/thunks/*</code> , <code>authSlice.ts</code>	The slice holds only state and reducers; thunks are small and testable
<code>ValidationPipe</code> (whitelist) + <code>omit</code>	<code>main.ts</code> , Prisma	No mass assignment, secrets do not reach the client
<code>.env</code> validated at startup, secrets ≥ 32 characters	<code>env.validation.ts</code>	Cannot start with an empty or weak secret
helmet, CORS by origin list	<code>main.ts</code>	Baseline protection of headers and cross-site requests
Single-flight refresh and <code>_retry</code>	<code>lib/axios.ts</code>	No races and no infinite loops
Safe <code>redirect_uri</code>	<code>isSafeRedirectPath</code>	Protection against open redirect
Uniform error format without stack traces	<code>AllExceptionsFilter</code>	The client does not see server internals

9. Known gaps and recommendations

This is a backlog for the team: the normal list of improvements for any project at this stage. Every item was verified against the current code; each gives the file, the substance and a proposed fix. Items are ordered by importance. First, what has already been fixed since the previous revision of these notes.

Resolved since the previous revision

Finding	What changed
Reset token not single-use	The token carries <code>passwordVersion</code> ; <code>resetForgottenPassword</code> claims it with a conditional <code>updateMany</code> , a replay gets 400
No <code>try/catch</code> around <code>verifyAsync</code> in <code>resetPassword</code>	Wrapped; an expired or malformed token gives 400 "Session is Expired"
<code>ResetPasswordDto</code> had no length rules	<code>@MinLength(8) @MaxLength(72)</code> (with a spec); the page inputs got <code>minLength/maxLength</code> too
Reset left other sessions alive	It now revokes all sessions of the user (the <code>currentSessionId</code> parameter is gone)
Optional <code>otp</code> in <code>ForgotPasswordDto</code>	Split: <code>ForgotPasswordDto (email)</code> and <code>VerifyForgotPasswordOtpDto (email + required otp)</code>
No refresh-token reuse detection	A stale token that no longer matches the hash revokes the session before the 401
<code>Verify2FaOtpPage</code> watched the wrong loading flag	It reads <code>isVerify2FaLoginOtpLoading</code>
<code>lastActiveAt</code> changed only on refresh	<code>JwtStrategy.validate</code> touches it at most once per 60 s
Expired sessions in the device list	<code>listSessions</code> filters <code>expiresAt > now</code>
Unescaped user name in emails	<code>escapeHtml(userName)</code> in the 2FA and reset emails
Typos <code>clearStoradge</code> , <code>commision-payouts</code>	Renamed to <code>clearStorage</code> , <code>commission-payouts</code>
Unused <code>create-auth.dto.ts</code> / <code>update-auth.dto.ts</code>	Deleted; DTOs reorganised behind a <code>dto/index.ts</code> barrel

WARNING Warning 1. No role-based authorization (`userType` is not checked)

Where: `jwt.strategy.ts`, `routes/index.tsx`, `authSlice.ts`

What the code does: The backend puts `userType` into `request.user`, but there is no `RolesGuard/@Roles(...)` anywhere. The frontend does not check the role either: after a successful login any user lands on `/dashboard`. In `authSlice`, `userType` is hard-coded to `"PLATFORM_ADMIN"` everywhere regardless of the response, and the `localStorage` value (`USER_TYPE`) is never read for access decisions. In other words, **login is implemented, but permission separation is not**: for example, a `CUSTOMER` user with a valid password can open the admin console. The practical risk is small today: the other dashboard pages are stubs, and there are no admin endpoints on the backend yet. But as soon as they appear, the missing `RolesGuard` becomes a hole.

Proposed fix: Add `@Roles('PLATFORM_ADMIN') + RolesGuard` (reads `request.user.userType`) on the backend for all admin routes; on the frontend, a `ProtectedRoute` that checks `user.userType`. Hiding buttons on the frontend is not enough: real protection exists only on the server.

WARNING Warning 2. Seed creates an administrator with a simple fixed password

Where: `backend/prisma/seed.ts`

What the code does: For `PLATFORM_ADMIN` the email and a very simple password (`12345678`) are hard-coded; on a repeated seed run it is **overwritten** (`update: { passwordHash }`).

Proposed fix: Take the password from an environment variable, do not run seed against a production database without changing the password, and enable 2FA on such an account.

IDEA Idea 3. Tokens are stored in localStorage**Where:** `lib/storage.ts`**What the code does:** Any JavaScript on the page (for example injected through XSS or a dependency) can read both tokens, including the 30-day refresh token. In addition, `storage.clearStorage()` calls `localStorage.clear()` and wipes **all** keys of the origin, not only the auth ones.**Proposed fix:** For production, the refresh token is often kept in a cookie with the `HttpOnly; Secure; SameSite` flags and the access token in memory. This reworks both the backend (issuing the cookie; `credentials: true` is already enabled in CORS) and the frontend. At the current stage localStorage is an acceptable trade-off; what matters is that the team understands it. Meanwhile, `clearStorage` can remove only the three known keys.**IDEA Idea 4. Refresh-token reuse: no alerting, and a cross-tab race****Where:** `auth.service.ts` → `refresh`, `lib/axios.ts`**What the code does:** Reuse detection exists (a stale token revokes the session), but it only revokes: there is no alert or notification, and the session family is not tracked. Also, single-flight refresh works only within one tab: if two tabs refresh at the same moment with the same old token, the second one looks like a replay and the session is revoked, so the user is signed out.**Proposed fix:** Log or notify on reuse; optionally allow a short grace window for the previous token, or coordinate tabs (for example with the Web Locks API or a `storage` event) so that only one tab refreshes.**IDEA Idea 5. One-time codes and the 2FA token: small leftovers****Where:** `auth.service.ts` → `forgotPassword`, `twoFactorCreateOtpToken`**What the code does:** A new forgot-password request does not invalidate older unused codes (login and toggle codes do get invalidated), so several codes can be valid at once, each limited by 5 attempts and the 5-minute TTL. The `twoFactorToken` of the login flow is not single-use either: within its 15 minutes it can be presented repeatedly, although it is useless without a correct emailed code.**Proposed fix:** Invalidate earlier unused reset codes on each new request; optionally bind the `twoFactorToken` to the current code (for example by a code id in the payload) so that it dies once used.**IDEA Idea 6. Minor findings****Where:** various files**What the code does:** • Changing the email in `PUT /users/me` requires neither a password nor mailbox confirmation. • `fetchMe.pending` sets `user = null`, so the profile disappears from the UI during any refetch. • The `twoFactorSecret` field and the `Role` type on the frontend are not used yet. • The folder name `pages/auth/forgot-pasword` is still misspelled (it is referenced by `routeModules.forgotPassword`), harmless but easier to fix while the codebase is small.**Proposed fix:** Work through these as the project grows.

9.1. What true multi-tenancy requires

- A `Tenant` table (store/organization) and a link between user and tenant (including many-to-many, if one person works in several stores).
- `tenantId` in the session and/or in the access token, and also in **every** domain table (products, orders...).
- A server-side filter by `tenantId` in all queries (for example via a Prisma extension or a base repository), so that forgetting `where: { tenantId }` is impossible.
- Roles inside a tenant (owner, manager...) and `RolesGuard/PermissionsGuard`; the platform admin role is kept separate.
- Tests for cross-tenant leaks: a user of tenant A must not see or modify the data of tenant B.

10. Checklist and review questions

10.1. Checklist: 15 key points

1. **Authentication ≠ authorization.** Login (who the user is) is implemented; permissions (what the user may do) are not yet.
2. The access token is short-lived (15 min) and sent with every request. The refresh token is long-lived (30 days) and used only for renewal.
3. A JWT is **signed but not encrypted**: the payload is readable by anyone, so secrets must not be stored in it.
4. Every session has a row in `user_sessions`; the access token contains `sid`, a reference to it.
5. `JwtStrategy.validate` checks the session in the DB on every request, which is why logout, password change and blocking take effect immediately.
6. The DB holds the **HMAC hash** of the refresh token, not the token itself; comparison uses `timingSafeEqual`.
7. Refresh is **rotated**, and **reuse is detected**: an old refresh token no longer works, and presenting it revokes the whole session.
8. On the frontend there is one refresh for all parallel requests (`pendingRefresh`); otherwise a race breaks the login.
9. `refreshClient` is separate from `Axios` so that the refresh does not loop on itself.
10. 2FA uses two "proofs": `twoFactorToken` (password verified) + the emailed code (control of the mailbox).
11. Codes: 6 digits, bcrypt hash, 5 minutes, 5 attempts, single-use; `purpose` separates the flows.
12. `login` and `forgot-password` do not reveal whether an email exists; `login` also equalizes response time.
13. Password change/reset revokes sessions (all others / all). The reset token is **single-use**: it carries `passwordVersion`, which stops matching once the password changes.
14. The frontend makes no security decisions: any check in the browser is only a convenience, real protection is on the server.
15. Tokens in `localStorage` are convenient but vulnerable to XSS; roles are not checked; the seed password is weak. These are the main backlog items.

10.2. Onboarding and review questions

Useful for onboarding new team members and for code review of authentication changes: try answering aloud first, then compare with the answers.

Question	Answer
1. Why two tokens rather than one long-lived token?	The access token goes out with every request and therefore lives briefly, so a stolen one expires quickly. The refresh token is used rarely and only on <code>/auth/refresh</code> , so it can live long and is still revocable through its session.
2. How can an access token be "revoked" if a JWT is stateless by itself?	The token contains <code>sid</code> . On every request <code>JwtStrategy</code> reads the session from the DB; if <code>revokedAt</code> is set or the session has expired, the result is 401. This adds a DB query but gives instant revocation.
3. What is stored in <code>user_sessions.refreshTokenHash</code> and why not the token itself?	An HMAC-SHA256 of the token with a server secret. If the DB leaks, a working refresh token cannot be derived from the hash.
4. What happens if two requests get a 401 at the same time?	Thanks to <code>pendingRefresh</code> , only one <code>/auth/refresh</code> goes to the server; both requests wait for the new access token and are retried. Without it, the second refresh with an already used token would fail and sign the user out.

Question	Answer
5. Why does <code>login</code> call <code>bcrypt.compare</code> even for a non-existent email?	So that response time is the same and it is impossible to tell by speed whether an email is registered.
6. How does <code>twoFactorToken</code> differ from the emailed code?	The token is a server-signed "password verified, waiting for the second factor"; the code is proof of access to the mailbox. Both are needed.
7. Why do 2FA codes have a <code>purpose</code> field?	So that a login code cannot be used to enable/disable 2FA, and vice versa.
8. Why does <code>forgot-password</code> always answer "success"?	So that it is impossible to find out by brute force which emails exist in the system.
9. What happens to other devices on a password change? And on a reset?	Change: all sessions except the current one are revoked. Reset: all sessions are revoked.
10. How is the password-reset token made single-use without a database table?	The token carries <code>passwordVersion</code> , the user's <code>password_updated_at</code> at issue time. <code>resetForgottenPassword</code> does <code>updateMany</code> with <code>where passwordUpdatedAt = passwordVersion</code> and sets a new <code>passwordUpdatedAt</code> . The first request updates one row; a replay or a parallel second request updates zero rows and gets 400.
11. What happens when an already used refresh token is presented, and what is the caveat?	The hash does not match, so the session is revoked and the client gets 401: the user has to sign in again. Caveat: two tabs refreshing simultaneously with the same old token can trigger this for an honest user, because single-flight works only within one tab.
12. What do <code>whitelist: true + forbidNonWhitelisted: true</code> do and what do they protect against?	They strip/forbid fields not described in the DTO. This protects against mass assignment: a client cannot send <code>userType</code> or <code>status</code> in <code>PUT /users/me</code> .
13. Why does <code>isSafeRedirectPath</code> forbid paths starting with <code>//</code>?	The browser treats <code>//evil.com</code> as the address of another site; without the check, <code>?redirect_uri=</code> would let an attacker send a user to a foreign page after login (open redirect).
14. Name the three main weak points of the current authorization.	No role checks (<code>userType</code>); tokens in <code>localStorage</code> (and <code>clearStorage</code> wipes everything); the fixed seed admin password. Smaller: the cross-tab refresh race, older reset codes not invalidated, the 2FA token not being single-use.
15. What is missing from the authentication layer for multi-tenancy?	A tenant entity, <code>tenantId</code> in the session/token and in the data, server-side filtering by tenant, roles inside a tenant and guards.

Appendices

A. How to observe the flows

- **DevTools → Network.** Open the tab, sign in to the app and look at the `login`, `me`, `refresh` requests: Payload shows what the browser sent, Preview shows what the server answered.
- **DevTools → Application → Local Storage.** Find `ACCESS_TOKEN`, `REFRESH_TOKEN`, `USER_TYPE`. To decode the payload: `JSON.parse(atob(token.split('.')[1].replace(/-/g, '+').replace(/_/g, '/')))` in the console.
- **Check the automatic refresh.** Set `JWT_ACCESS_EXPIRES_IN="1m"` in `backend/.env`, sign in and after a minute perform any action: Network shows 401 → `/auth/refresh` → the request repeated with 200.
- **Check revocation.** Sign in from two browsers, revoke the second session in "Active sessions"; in the second browser the next request returns 401 and the refresh fails.
- **Prisma Studio** (`npm run prisma:studio` in `backend/`): watch rows appear in `user_sessions` and `two_factor_otp`s and how `revokedAt`, `attempts`, `verifiedAt` get filled in.

Running the project

```
# backend/
cp .env.example .env          # fill in DATABASE_URL, 4 secrets (>=32 characters), RESEND_*
npm install
npm run prisma:deploy          # apply migrations (or prisma:migrate in development)
npm run db:seed                # create the administrator (see Warning 2 in part 9)
npm run start:dev              # http://localhost:4000

# platform-admin/
npm install
npm run dev                    # http://localhost:3002 (VITE_APP_API_BASE_URL, defaults to localhost:4000)
```

B. Authentication file map

File	Purpose
backend/prisma/schema.prisma	Models and enums
backend/src/main.ts	helmet, ValidationPipe, error filter, CORS
backend/src/app.module.ts	Config, Throttler, Passport, modules
backend/src/config/env.validation.ts	.env validation at startup
backend/src/auth/auth.controller.ts	/auth/* endpoints
backend/src/auth/auth.service.ts	All logic for login, tokens, sessions, 2FA, password reset
backend/src/auth/strategies/jwt.strategy.ts	Access-token and session verification
backend/src/auth/dto/* (index.ts barrel), types/*	DTOs (LoginDto, ForgotPasswordDto, VerifyForgotPasswordOtpDto, ResetPasswordDto, TwoFactor*) and token/response types
backend/src/**/*spec.ts, backend/vitest.config.mts, vitest.config.e2e.mts	Backend unit/e2e tests and their vitest configs
backend/src/common/guards/jwt-auth.guard.ts	Guard for protected routes
backend/src/common/decorators/*	@CurrentUser(), @ParseUserAgent()
backend/src/common/filters/all-exceptions.filter.ts	Uniform error format
backend/src/users/*	Profile, password change, password reset
backend/src/email/email.service.ts	Emails with codes (Resend)
backend/prisma/seed.ts	Administrator creation
platform-admin/src/lib/axios.ts	Interceptors and automatic token refresh
platform-admin/src/lib/storage.ts	localStorage
platform-admin/src/api/auth.ts	API calls
platform-admin/src/store/auth/authSlice.ts, auth.helpers.ts	Redux state and reducers
platform-admin/src/store/auth/thunks/*	Thunks: session, profile, twoFactor, sessions, forgotPassword
platform-admin/src/lib/validators.ts	validateEmail, validatePassword
platform-admin/src/routes/index.tsx, components/ProtectedRoute.tsx, layouts/AuthLayout.tsx	Routes and page protection
platform-admin/src/pages/auth/login/* (LoginPage, validateLogin, hooks/)	Login screen: a thin page plus hooks
platform-admin/src/pages/auth/forgot-password/* (sic)	Forgot-password screen and its hooks
platform-admin/src/pages/auth/*Page.tsx	2FA, OTP and reset-password screens
platform-admin/src/pages/dashboard/account/*	Profile, password, 2FA, sessions

C. For more detail

The repository already contains earlier materials: [AUTH_FLOW.md](#) (a walkthrough with diagrams and "On the wire" JSON examples) and [AUTH_FLOW_RU.pdf](#) (a Russian walkthrough with screenshots of the Network tab). The backend now compiles to CommonJS (`"type": "module"` removed, imports without `.js` suffixes, `prisma moduleFormat = "cjs"`, `void bootstrap()` in `main.ts`). The repository also has [README.md](#) and [AUTH_SEQUENCE.md](#). These notes were rebuilt from the current code (commit [c1d2eb9](#)) as team documentation: a glossary of concepts, a layer-by-layer breakdown, flows, a backlog of known gaps and review questions.